

NEXUM SUUM | 24-HOUR COOLING TOWER SOP

This Standard Operating Procedure (SOP) governs the 24-hour operational and safety checks for Cooling Towers made by Marley, EVAPCO, BAC, and SPX. Tasks are divided into shift-specific duties to support optimal system performance, safety, and compliance with manufacturer and ASHRAE 188 guidelines.

AFTERNOON SHIFT: 2:00 PM - 10:00 PM

- Recheck fan motor amp draw and condenser approach temps.
- Clean strainers and confirm spray nozzle performance.
- Verify chemical dosing levels (biocide, scale inhibitors).
- Check temperature differential from inlet to outlet.
- Test float switch and overflow protection.
- Log system conductivity and total dissolved solids (TDS).

NIGHT SHIFT: 10:00 PM - 6:00

AM

- Verify fan cycling logic based on outside air temperature.
- Inspect gear reducers for oil levels and shaft alignment.
- Check walkway lighting and railings for safety compliance.
- Review alarms and auto shutdowns in building automation.
- Drain water sample for overnight chemistry analysis.
- Confirm vibration readings and start/stop logs from night cycle.

FINAL 24-HOUR INSPECTION CHECKLIST

Task	Yes / No
All chemical levels logged and verified	
Fan motor amps within safe range	
Blowdown and overflow controls inspected	
Fill media and eliminators clear of obstructions	
Vibration readings logged correctly	
Overnight alarms acknowledged and documented	
Heating Bed unclogged	